

Project Executable Files

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	3 Marks

This section covers submission of all executable and deployable files for the project, so an evaluator can run or deploy the solution using the files and instructions below.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes — README.md + setup.md
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	N/A — no database; interaction history is CSV-based only
5	Environment configuration file (.env.example or similar)	Yes — .env.template (see Known Issues: a couple of variable names in it are stale)
6	Deployed application URL (if hosted)	Yes — see Step 3 (public deployment runs in a reduced mode; see remarks)
7	APK / Executable binary (if applicable for mobile/desktop apps)	N/A — web application only
8	Dockerfile / Containerization config (if applicable)	N/A — a Docker/HF Spaces path was evaluated and consciously not pursued (see DEPLOYMENT_NOTES.md)
9	Test files and test results	Yes — tests/test_e2e_ui_flow.py (10/10 passing), VALIDATION_REPORT.md, PERFORMANCE_TESTING.md
10	Demo video or walkthrough (if required)	Not yet recorded — to be added before final submission

Step 2: File / Folder Structure

Top-level layout of the repository:

app.py	- Streamlit entry point
src/	
inference/	- BiLSTM/BERT predictors, cached loaders, mixed-emotion,
schema	
generation/	- prompt builder, Gemini/OpenRouter clients, template
fallback	
orchestration/	- analysis_pipeline.py, decision_engine.py,
session_history.py	
persistence/	- csv_logger.py
preprocessing/	- dataset_builder, label_mapping, text_cleaning,
tokenization	
models/	- bilstm.py, focal_loss.py (model definitions)
ui/	- one file per UI concern (toggles, panels, dashboard,
sidebar)	
scripts/	- training scripts, demo scripts, integration checks
analysis/	- validation predictions (CSV) +
decision_signals_report.txt	

tests/	- test_e2e_ui_flow.py (10-test Streamlit AppTest suite)
envtesting/	- TensorFlow/PyTorch process-isolation tests
models/	- trained weights (gitignored; BiLSTM artifacts
committed,	BERT weights hosted on Hugging Face Hub)
data/, logs/	- processed datasets, opt-in interaction CSV log
1.-8. [numbered folders]	- this college submission paperwork

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://emotionalgemini.streamlit.app/ — runs in a deliberately reduced mode (BiLSTM-only, template responses), not the full dual-model + real-AI pipeline. See Known Issues for why.
Login Credentials (Demo)	Not applicable — the app has no authentication; it's open access.
Platform / Hosting Provider	Streamlit Community Cloud
Repository Link	https://github.com/weeb-with-net/emotion-detection-engine-lsa
Demo Video Link	Not yet recorded

Step 4: Run Instructions

Full instructions are in `setup.md`; summarized here:

```
1. git clone https://github.com/weeb-with-net/emotion-detection-engine-lsa.git
2. py -3.12 -m venv .venv      (Python 3.12 required - 3.14 conflicts with TensorFlow 2.16.1)
3. Activate the virtual environment
4. (If using GPU) pip install torch torchvision torchaudio --index-url
https://download.pytorch.org/whl/cu128
5. pip install -r requirements.txt
6. Copy .env.template to .env and fill in GEMINI_API_KEY and/or OPENROUTER_API_KEY
7. streamlit run app.py
```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Public deployment runs BiLSTM-only with template responses, not the full dual-model + real-LLM pipeline.	Intentional: an unresolved OpenRouter-call hang occurs specifically on Streamlit Cloud, worked around via <code>DISABLE_BERT</code> / <code>DISABLE_AI_RESPONSE</code> rather than fixed. Full behavior is verified locally (RTX 5070 Ti) — see <code>VALIDATION_REPORT.md</code> and <code>PERFORMANCE_TESTING.md</code> .
2	<code>.env.template</code> references a stale variable name (<code>GOOGLE_API_KEY</code>) and a typo'd model path (<code>models/bltstm</code>) that don't match the current code (<code>GEMINI_API_KEY</code> , <code>models/bilstm</code>).	Cosmetic — <code>setup.md</code> gives the correct current instructions, so this doesn't block setup. Recommend updating the template before final submission.
3	The "CSV-based prediction" toggle is visible in the UI but disabled.	Intentional and clearly labeled "planned feature" in the UI itself — no backing retrieval pipeline exists yet.
4	The Analytics "Summary" tab renders empty.	Intentional — no scope was defined for it; held as a clean slot for a future dashboard-metrics pass.
5	<code>logs/interactions.csv</code> (opt-in interaction history) won't persist across restarts on non-persistent hosting disks.	Acceptable for a single demo/assessment session; documented tradeoff, not a defect.